Automated Flame Tracking and Dynamics Analysis

A Complete Computer Vision and Mathematical Modeling Pipeline

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Primary Motivation

Overarching Goal: To develop a fully automated, highly robust pipeline for tracking flame propagation in experimental videos and extracting mathematically rigorous flow dynamics.

- **High-Noise Environments:** Processing experimental combustion data requires discarding immense amounts of visual noise to isolate the true physical phenomenon.
- **Precision Tracking:** We require absolute precision in identifying the flame's spatial coordinates over time.
- **Smooth Mathematical Extraction:** Extracting velocity and acceleration via direct numerical differentiation of raw pixel data amplifies noise significantly. A comprehensive mathematical modeling phase is necessary to determine true flame dynamics.

Key Challenges in Experimental Image Processing

1. Illumination Variation & Background Motion

- Rapid changes in overall lighting cause global pixel intensity shifts, tricking standard background subtractors.
- **Reflection Artifacts:** Bright reflections on the pipe walls or background structural elements create false "flames."
- Left unmitigated, the tracker shifts its bounding box toward these reflections, drastically corrupting the positional dataset.

Key Challenges in Experimental Image Processing

2. Centroid Skewing & Occlusion

- Tracking the geometric centroid of a bounding box often fails when the flame shape fractures, stretches irregularly, or faces partial occlusion.
- **Solution:** By anchoring our tracking logic strictly to the **Maximum Intensity** pixel, the system remains locked onto the hottest, densest core of the combustion front, maintaining a reliable anchor regardless of shape distortion or occlusion.

To solve these challenges, we designed a two-phase architecture.

- **Phase 1 (Computer Vision):** Optimal background selection, Static Eraser Boolean masking, and max-intensity contour refinement.
- **Phase 2 (Mathematical Modeling):** Saturating exponential curve fitting, analytical dynamics derivation, and characteristic Star normalization.

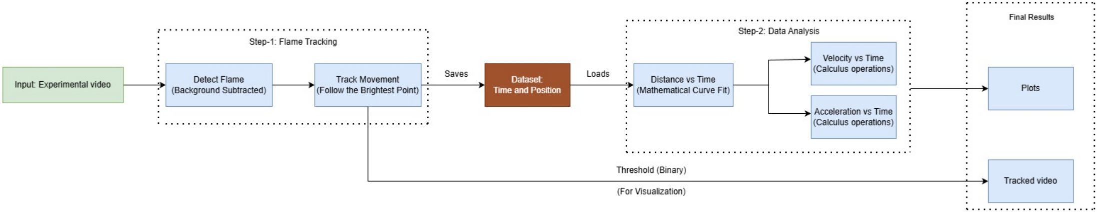

Fig A: High-level system architecture detailing the transition from raw experimental video to analytical plots.

Phase 1

Computer Vision

Step 1: System Initialization & Region Definition

Before analyzing motion, the environment must be strictly defined mathematically.

Spatial Zoning (Real-World to Pixels)

- Conversion constant established: ~ 4.85 Pixels/mm.
- **Pipe Zone:** The primary bounding box containing the experiment.
- **Ignition Zone:** Specifically monitored for the initial brightness spike signaling flame creation.
- **Close Zone:** Right-side boundaries requiring specialized morphology to prevent signal degradation.

Brute-Force ROI Masking

An initial logical mask (`roi_mask`) is generated to completely black out any pixels outside the `PIPE_ZONE`.

This completely prevents external laboratory motion from triggering the tracker.

Step 2: Intelligent Master Background Selection

A random first frame may contain an early spark or an anomalous reflection. We algorithmically guarantee the best background.

- **The Search:** The algorithm iterates through the first N (e.g., 20) frames of the dataset.
- **Targeted Analysis:** It crops each frame strictly to the `PIPE_ZONE`.
- **Intensity Scoring:** It calculates the total summed intensity of all pixels in this cropped region.
- **Selection:** The frame with the **absolute lowest total intensity** (the darkest, emptiest pipe) is locked in as the Master Background (`bg_gray`).

Why this matters

This ensures that when we subtract the background later, we are subtracting the true empty state of the pipe, maximizing the signal-to-noise ratio of the actual flame.

Step 3a: The Challenge of Pre-existing White Spots

Failure of Standard Background Subtraction

- Even in our algorithmically selected "darkest" frame, there are pre-existing white spots present inside the pipe from the start.
- When the dynamic flame front propagates, it physically crosses or passes through these specific spatial coordinates.
- Due to this direct overlap, standard absolute difference background subtraction is not enough to cleanly remove these spots.
- **The Tracking Hazard:** Because our tracker searches for maximum intensity, these residual bright spots act as false anchors, dragging the bounding box away from the true flame core.

Fig B: Inherent white spots present inside the pipe geometry prior to ignition (Phi_1p2).

Step 3b: The Static Eraser & Logical NOT Operation

Isolating the Anomalies

- A threshold (`STATIC_ERASER`) is applied to the Background Frame, mapping the exact coordinates of these pre-existing spots.

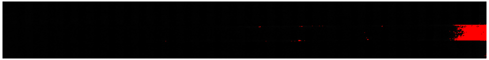

Fig C: The Keeper Mask. Red regions dictate where static white spots are permanently zeroed out (Phi_1p2).

The `bitwise_not` Operation

- We mathematically invert this map using `cv2.bitwise_not`.
- The problematic white spots become pure black (0), and the valid tracking environment becomes white (255).
- This mask is multiplied against all subsequent live frames. It physically deletes the static spots from the frame before the max-intensity search executes, ensuring undisturbed tracking.

Step 4: Absolute Difference & Zonal Morphology

With the environment mapped, we process the video frame-by-frame.

- **Gaussian Blur & Subtraction:** Both the current frame and Master Background are blurred (5×5) before computing the absolute difference to suppress micro-noise.
- **Binarization:** A threshold is applied to isolate the moving foreground.

Targeted Morphological Cleaning

- **Open Zone (`MORPH_OPEN`):** A heavy 20×20 kernel is applied over 40 iterations. This aggressively destroys small floating noise clusters caused by flickering illumination.
- **Close Zone (`MORPH_CLOSE`):** A 3×3 kernel is applied to bridge gaps and heal the binary shape of the primary flame body where it tends to fracture.

Step 5: HSV Enhancement & Cleansing

To extract maximum contrast for the tracker, the binary mask is mapped back to the original color frame.

V-Channel Contrast Boosting

- The isolated flame region is converted to HSV color space.
- The *Value* (Brightness) channel is artificially scaled up (`CONTRAST_FACTOR = 4.0`).
- This forces the flame core to peak intensity while suppressing the fringes.

Application of the Keeper Mask

- This enhanced grayscale image is multiplied by the *Keeper Mask* generated in Step 3.
- **Result:** The high-contrast flame remains, but the background reflections are mathematically zeroed out.
- The final clean image is thresholded for the tracking algorithm.

Step 6: Max Intensity & Contour Refinement

The tracking logic ensures the system follows the densest flame core, not random geometric noise.

1. Max Brightness Anchoring

- The algorithm scans a dynamic search area where `cv2.minMaxLoc` finds the absolute brightest pixel, centering a display box over the densest core.

2. Contour Refinement & Position

- `cv2.findContours` shrink-wraps the bounding box tightly to the active flame shape within that ROI.
- The final spatial position is calculated using the geometric **centroid** of this refined bounding box.

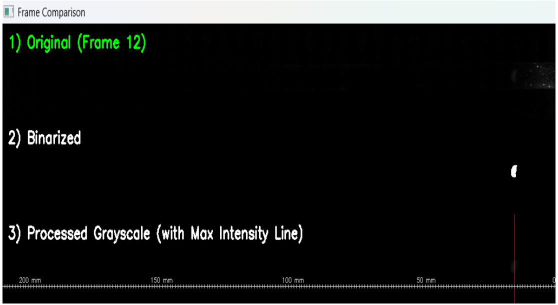

Fig D: Tracker anchors on max intensity (Phi_1p0).

Phase 2

Mathematical Modeling

Step 1: Data Ingestion & Temporal Scaling

Phase 1 reliably outputs a pristine CSV dataset of Frame ID, Timestamp (s), and Position (mm). Phase 2 converts this raw data into physical insight.

Temporal Normalization

- Real-world high-speed cameras record rapidly. To allow for 1:1 cross-experiment comparison, the raw timestamp arrays are multiplied by a scaling factor:

$$\text{Scaling Factor} = \frac{\text{TARGET\_DURATION (5.0s)}}{\text{Original Max Time}}$$

- This compresses or expands the dataset into a standardized 5-second analytical window without altering the shape of the data.

Step 2: Saturating Exponential Modeling

Why Curve Fitting?

- Direct numerical differentiation amplifies positional noise.
- We fit a mathematical model to represent the fundamental physical dynamics of a propagating, then decelerating, flame front.

Parameter Extraction

- Scipy's `curve_fit` (Least Squares) optimizes for k, t_0, C.
- A is locked relative to the empirical saturation boundary (average of the last 5 frames).

Position Model $P(t)$

$$P_{eq}(t) = A\left(1 - e^{-k(t-t_0)}\right) + C$$

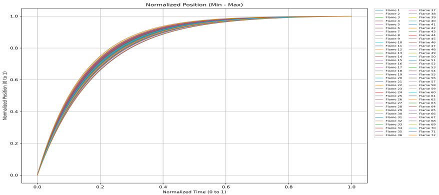

Fig E: Normalized position plot for `Phi_1p0_u_0p3_C001H001S0001`.

Step 3: Analytical Derivation of Dynamics

With a continuous mathematical function for position, motion dynamics are derived analytically, yielding smooth profiles devoid of experimental jitter.

Velocity Profile $V(t)$

$$V_{eq}(t) = \frac{d}{dt}P(t) = Ak\, e^{-k(t-t_0)}$$

This perfectly captures the peak initial velocity followed by continuous decay as the flame stabilizes.

Acceleration Profile $A_{cc}(t)$

$$A_{eq}(t) = \frac{d}{dt}V(t) = -Ak^2\, e^{-k(t-t_0)}$$

Yields a smooth, negative acceleration curve (deceleration) devoid of experimental high-frequency jitter.

Step 4: Characteristic "Star" Normalization

To generalize the findings across varied equivalence ratios or flow speeds, we project the data into a non-dimensional space using characteristic constants (T^*, P^*, V^*).

- **Characteristic Constants:** Identified as $T^* = 0.000261$, $P^* = 0.107$, and $V^* = 409.881$.
- We map standard limits to the 0-1 bounds (Min-Max scaling).
- The Star Normalization transforms the fitted curves into a universal comparative manifold:

Non-Dimensional Transform

$$P^\star(t^\star) = \frac{A}{P^*}\left(1 - e^{-k^\star(t^\star - t_0^\star)}\right) + \frac{C}{P^*}$$

(Where $t^\star = t/T^*$ and $k^\star$ is appropriately scaled)

Results

Results: Position Tracking & Saturating Exponential Fit

Positional Data Extraction

- The proposed CV pipeline successfully extracted spatial coordinates for the flame's Maximum Intensity.
- The empirical data shows a distinct rapid initial progression that gradually plateaus as the flame approaches the end of the pipe geometry.

Model Validation

- The 3-parameter saturating exponential model tightly bounds the raw data.
- By locking the A parameter to the observed saturation limit, the least-squares fit optimally captures the true progression curve.

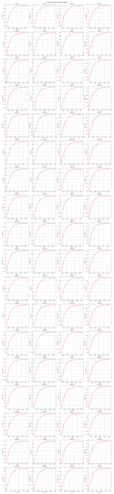

Fig F: Extracted position (dots) for each flames of `Phi_1p0_u_0p3_C001H001S0001`.

Results: Analytical Dynamics (Velocity & Acceleration)

Velocity Profile

- Derived analytically from the position model.
- Demonstrates peak velocity at the ignition phase, followed by a continuous exponential decay.

Acceleration Profile

- The second derivative reveals a smooth, purely negative acceleration (deceleration) curve.
- Confirms that the flame front experiences maximum resistance forces immediately after the initial explosive expansion.

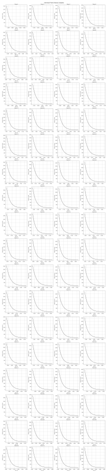

Fig G: Derived Velocity $Phi\_1p0\_u\_0p3$.

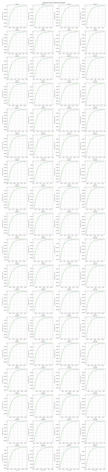

Fig H: Derived Acceleration $Phi\_1p0\_u\_0p3$.

Results: Automated Flame Tracking Validation

Visual Artifacts

- **Target Lock:** The tracker successfully maintains its anchor on the maximum intensity core throughout the entire run.
- **Scale Verification:** Scale bar accurately maps pixel coordinates to physical millimeters.
- **Data Export:** The extracted (t, P_{mm}) coordinates form the foundational dataset for Phase 2.

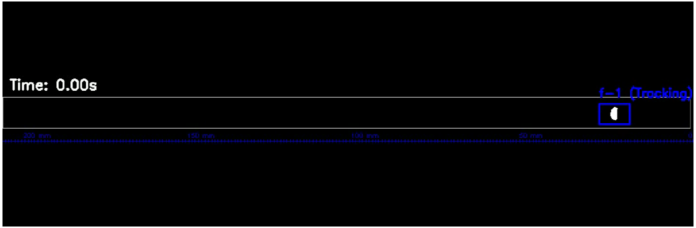

Fig I: Processed tracking output (Phi_1p0_u_0p3_C001H001S0001). (Click image to view video).

References

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Thank You

Questions & Discussion